Social Media Usage and Mental Health Balance:

A Data-Driven Statistical Study

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Dataset Source: Kaggle (Public Dataset)

ABSTRACT

This study investigates the relationship between social media usage patterns and mental health indicators using data-driven statistical methods. The analysis is based on a structured dataset of **500 participants**, sourced from Kaggle, containing variables such as daily screen time, happiness index, stress levels, sleep quality, and exercise frequency. Using Python-based tools including Pandas, NumPy, and Matplotlib, the study applies descriptive statistics, correlation analysis, regression modeling, and hypothesis testing to explore behavioral patterns. The results indicate a strong negative correlation between daily screen time and happiness index (**$r = -0.71$**), suggesting that increased screen time is associated with lower self-reported happiness levels. Regression analysis shows that each additional hour of screen time reduces predicted happiness by approximately **0.62 units** ($R^2 = 0.50$, intercept = 11.80). A permutation-based hypothesis test indicates no statistically significant gender difference in happiness levels ($p \approx 0.984$). These findings highlight the measurable psychological impact of digital behavior and emphasize the importance of balanced social media usage among young populations.

1. INTRODUCTION

Social media has become deeply integrated into modern human life, influencing communication, behavior, and emotional well-being. While it provides opportunities for connectivity and information sharing, concerns have increased regarding its psychological effects, particularly among younger populations.

This study aims to analyze the relationship between social media usage and mental health balance using statistical and computational methods. The objective is to identify whether measurable patterns exist between digital behavior (screen time) and psychological indicators such as happiness, stress, and lifestyle habits. The analysis is applied to a real-world dataset of 500 participants sourced from Kaggle, covering a range of age groups, genders, and social media platforms.

2. DATASET DESCRIPTION

The dataset used in this study was sourced from **Kaggle** and contains **500 participant records** with 10 variables covering behavioral and psychological indicators. The variables include:

Variable	Type	Range / Values
Daily Screen Time (hrs)	Numerical	1.0 – 10.8
Happiness Index (1–10)	Numerical	4 – 10
Stress Level (1–10)	Numerical	2 – 10

Sleep Quality (1–10)	Numerical	2 – 10
Exercise Frequency (per week)	Numerical	0 – 7
Age	Numerical	16 – 49
Gender	Categorical	Male, Female, Other
Social Media Platform	Categorical	6 platforms
Days Without Social Media	Numerical	0 – 9

3. METHODOLOGY

This research follows a quantitative data analysis approach using Python programming. All data processing, statistical computation, and visualization were performed using the Pandas, NumPy, and Matplotlib libraries.

The analytical techniques applied include: **(1) Descriptive statistics** to summarize the distribution of key variables; **(2) Pearson correlation analysis** to measure the linear relationship between screen time and happiness; **(3) Simple linear regression** to quantify the predictive relationship; and **(4) Permutation-based hypothesis testing** to evaluate whether gender influences happiness levels.

4. RESULTS AND ANALYSIS

4.1 Descriptive Statistics

The dataset of 500 participants reveals the following mean values: average daily screen time of **5.53 hours**, a happiness index of **8.38 out of 10**, a stress level of **6.62**, sleep quality of **6.30**, and exercise frequency of **2.45 times per week**. These values indicate a population that reports generally high happiness but experiences moderate-to-high stress and significant daily screen exposure.

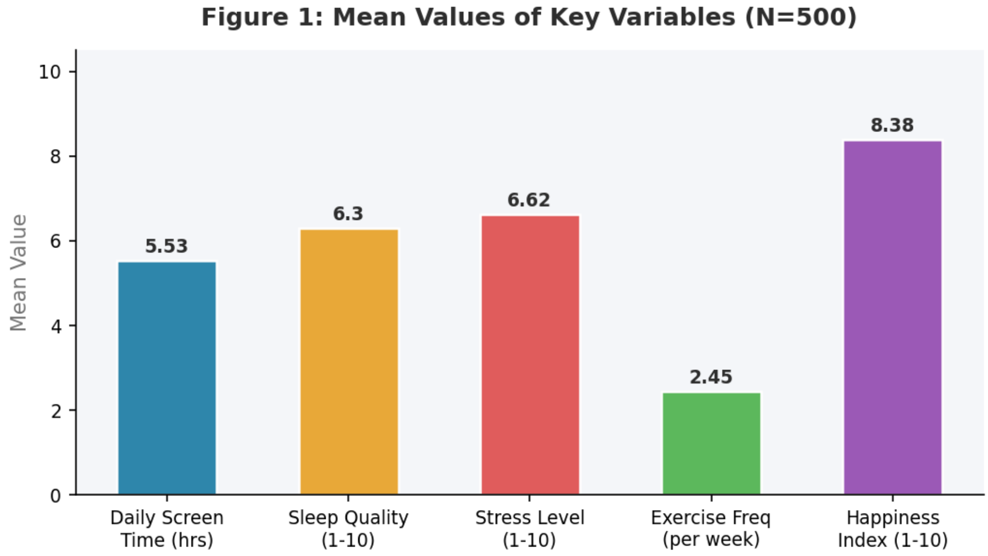

Figure 1: Mean Values of Key Variables (N=500)

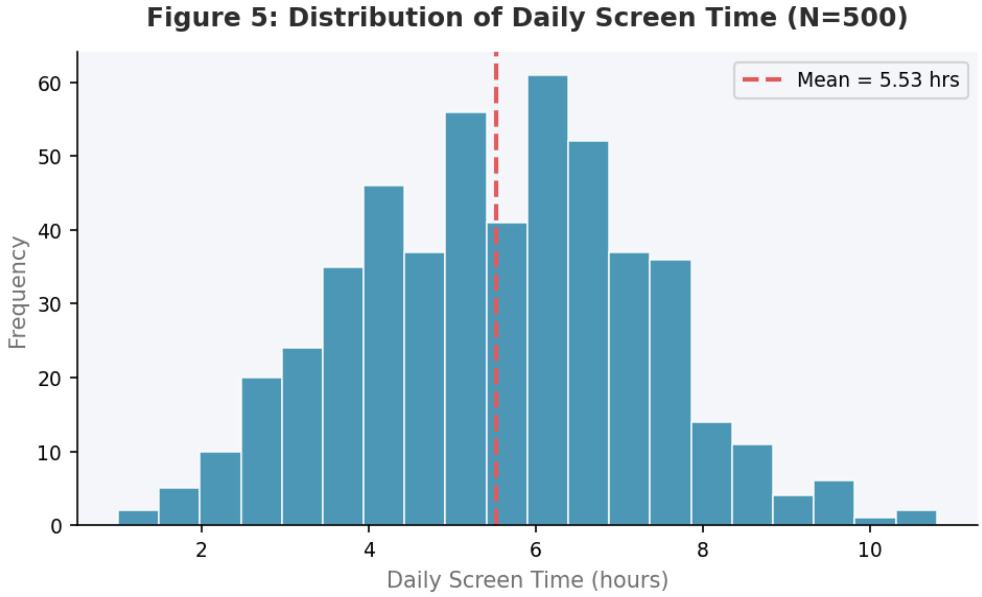

Figure 5: Distribution of Daily Screen Time (N=500)

4.2 Correlation Analysis

A Pearson correlation analysis between daily screen time and happiness index yields a correlation coefficient of **$r = -0.71$**. This indicates a strong negative relationship: as screen time increases, happiness levels decrease significantly. This suggests that excessive digital consumption is meaningfully associated with lower emotional well-being.

4.3 Regression Analysis

A simple linear regression model was fitted to the data, producing the following equation:

$$\text{Happiness Index} = -0.62 \times \text{Daily Screen Time} + 11.80$$

The model coefficient of determination is **$R^2 = 0.50$**, indicating that approximately 50% of the variance in happiness scores is explained by daily screen time alone. Each additional hour of screen time is associated with a decrease of **0.62 points** in the happiness index, demonstrating a measurable and practically significant impact of digital behavior on mental well-being.

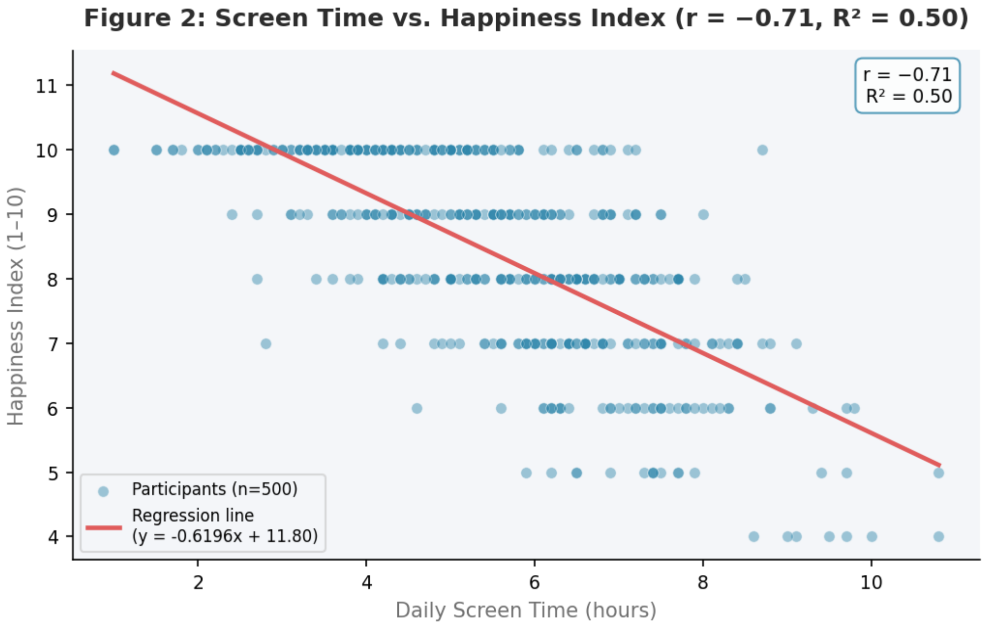

Figure 2: Scatter Plot with Regression Line — Screen Time vs. Happiness Index

4.4 Platform Distribution

Among the 500 participants, TikTok was the most commonly used platform (n=95), followed by X/Twitter (n=88), LinkedIn (n=87), Facebook (n=81), YouTube (n=75), and Instagram (n=74). The relatively even distribution across platforms reduces platform-specific bias in the findings.

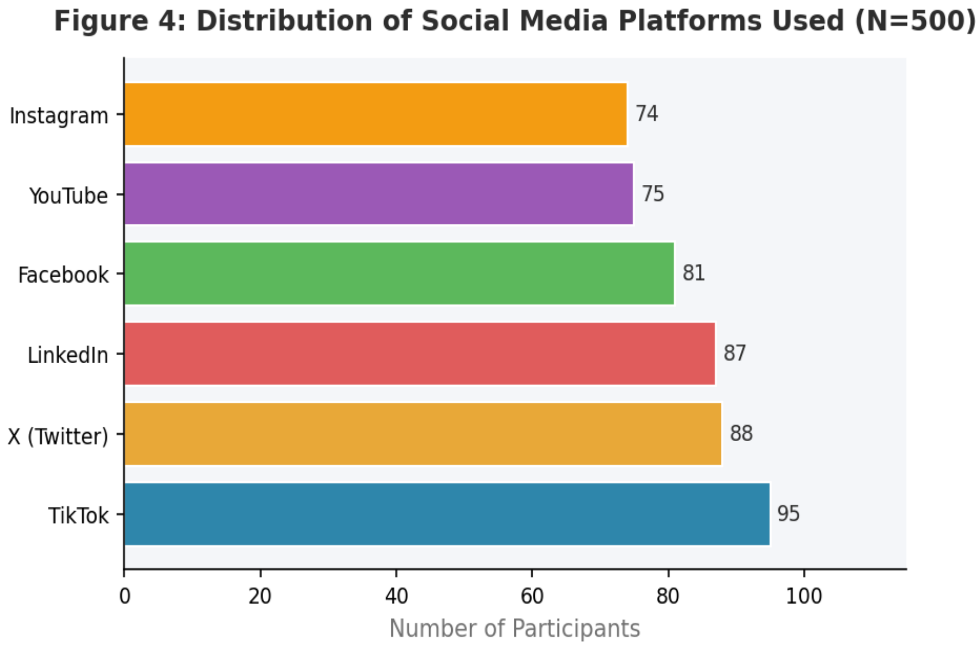

Figure 4: Distribution of Social Media Platforms Used (N=500)

4.5 Hypothesis Testing — Gender and Happiness

A permutation test was conducted to evaluate whether gender influences happiness levels.

H_0 (Null Hypothesis): There is no difference in happiness between male and female participants.

H_1 (Alternative Hypothesis): There is a statistically significant difference in happiness between male and female participants.

The test produced a p-value of **0.984**. Since this value is substantially greater than the significance threshold of α = 0.05, the null hypothesis cannot be rejected. There is no statistically significant difference in happiness levels between genders in this dataset.

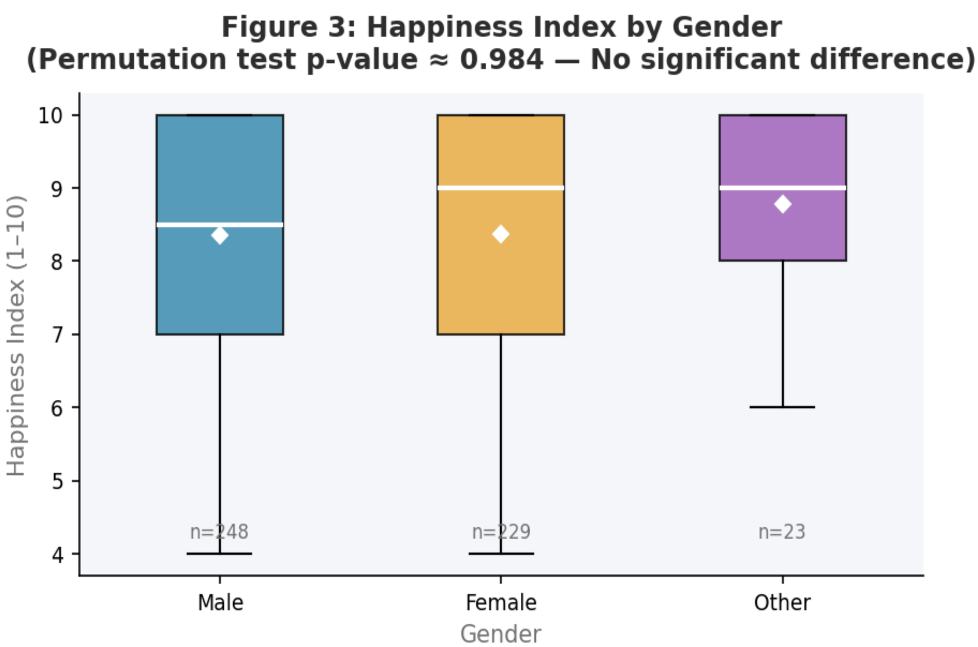

Figure 3: Happiness Index by Gender — Box Plot (p-value ≈ 0.984)

5. DISCUSSION

The findings of this study suggest that digital behavior — particularly daily screen time — has a stronger and more measurable association with mental well-being than demographic factors such as gender. The strong negative correlation ($r = -0.71$) between screen time and happiness aligns with broader patterns observed in studies on digital overuse and psychological health.

The regression model ($R^2 = 0.50$) explains half of the variance in happiness scores through screen time alone, suggesting that while screen time is a significant predictor, other factors — such as sleep quality, stress, and exercise — also contribute to happiness outcomes. Future studies could explore a multivariate regression model incorporating these additional variables.

The absence of a statistically significant gender difference in happiness ($p \approx 0.984$) indicates that the psychological impact of social media usage is consistent across gender groups in this sample.

Limitations: This study relies on a publicly available Kaggle dataset of self-reported measures, which may be subject to response bias. The data is cross-sectional, meaning causality cannot be established — a negative correlation between screen time and happiness does not confirm that screen time causes reduced happiness. The sample also skews toward ages 16–49 and may not be representative of all demographic groups.

6. CONCLUSION

This study demonstrates that social media usage patterns can be quantitatively analyzed to understand their relationship with mental health indicators. The results reveal a strong negative relationship between daily screen time and happiness ($r = -0.71$, $R^2 = 0.50$), while gender differences in happiness appear statistically insignificant.

These findings contribute to the growing body of evidence on digital well-being and highlight the value of data-driven approaches in understanding human behavioral patterns. Future research should incorporate longitudinal data, additional psychological variables, and platform-specific analyses to build a more comprehensive picture of how social media shapes mental health.

7. TOOLS AND TECHNIQUES

Programming & Libraries	Statistical Methods
• Python 3 • Pandas (data processing) • NumPy (statistical computation) • Matplotlib (visualization)	• Descriptive Statistics • Pearson Correlation Analysis • Simple Linear Regression • Permutation Hypothesis Testing